Team RNG

Factory Energy Cost Decomposition and Savings Agent

Our approach, for mentor review

HCLTech Future Ready AI Engineer Hackathon • IIIT Delhi • 11 October 2026 • github.com/silverbolt08/RNG

1. What we are building

A factory's electricity bill is 15 percent higher than last quarter. Our system attributes that rise to tariff, volume, product mix, weather and controllable waste, in that order, and only then recommends actions with costed savings, CO2e and payback. The five causes add up to the total by construction, not by adjustment.

The governing principle for every design choice below: **the language model classifies the request and writes the prose. It never computes a number and never decides what is wasteful.** Numbers come from code, and every number we print can be traced to the tool that produced it.

2. System architecture

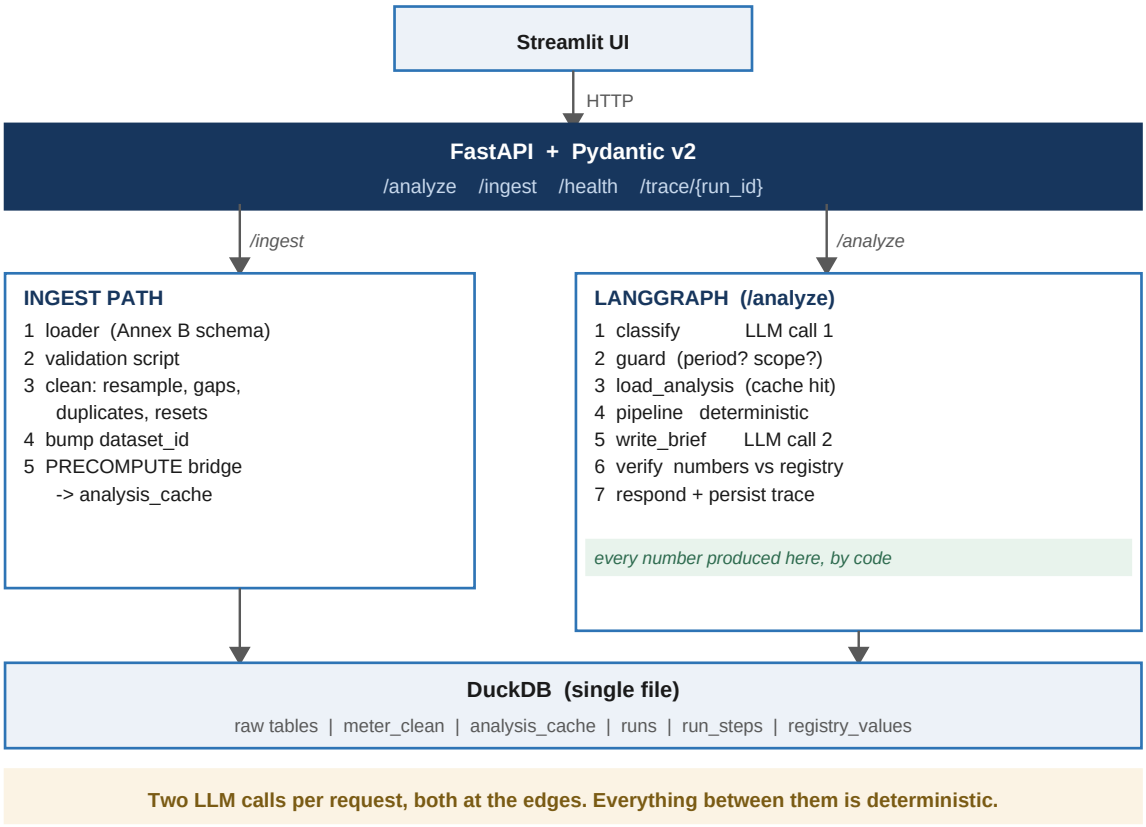

Figure 1 — Request flow. The UI calls the API over HTTP rather than importing the modules, so our demo exercises the same contract the judges test.

Analysis is computed once when data lands and cached against the dataset version, so questions are answered from a warm cache. That keeps p50 and p95 latency low, and it makes the late data drop cheap to handle: a new ingest produces a new version, and "what changed" is the difference between two cached bridges, cause by cause.

3. Design decisions, and why

The brief rewards the simplest architecture that meets the requirements. These are the choices we made and the reasoning we will defend.

Decision	What we chose	Why
Request routing	LLM classifies intent into structured JSON, then deterministic code runs a fixed pipeline per answer type. Two LLM calls per request.	There are six known question types. An agentic tool-picking loop would add variable latency and an unbounded call count without improving accuracy on a closed set, and it makes traces harder to read.
Attribution method	Shapley values over the four explainable causes; waste is the residual.	Sequential attribution gives a different answer depending on the order you process causes in. Shapley averages over all orderings, so no cause is advantaged by its position.
Number safety	Every tool result enters a number registry. The brief is validated against it after generation; on a mismatch we regenerate once, then fall back to a slot-filled template.	Makes a hallucinated figure structurally unable to reach the response, while keeping the brief readable. The worst case is stilted prose, never a wrong number.
Attribution order	Enforced by function signature, not by prompt wording.	detect_leaks() cannot be called without the completed bridge, so nothing can be named as waste before tariff, volume, mix and weather have been removed.
Data store	DuckDB, single file.	Analytical queries over roughly 158,000 meter rows, reads Parquet natively, and the brief warns against loading time series into SQLite to satisfy the rule.
Compute timing	Precompute on ingest, serve from cache.	Latency is a reported metric, and judges ingest data and then immediately ask questions. We pay the cost once, where it is expected.
Agent count	One classifier, one writer. No further agents.	The decomposition, idle-load detection, savings and ranking are all deterministic. Additional agents would add failure modes without adding capability.
LLM	Gemini 3.5 Flash Lite by default, selected by LLM_PROVIDER and LLM_MODEL; MOCK_LLM runs the full pipeline with no model at all.	Only two calls per request, so a fast low-cost model is sufficient. The switch lets us move to Gemini 3.8 Flash if brief quality needs it, and MOCK_LLM keeps us working if the network does not.

4. How we split the 15 percent

The bill is roughly the sum of energy multiplied by price. Several drivers moved at once, and they interact, so the cross terms have to be assigned to someone. The assignment is not unique. Taking a simplified two-factor case where energy rose from 1,000 to 1,100 and price from 10.00 to 10.45, a total rise of 1,495:

Order of attribution	Price effect	Volume effect	Total
Price measured first	450	1,045	1,495
Volume measured first	495	1,000	1,495

Both add up. Both are defensible. The price effect differs by 45 purely because of processing order, and with four causes there are 24 orderings. We therefore use Shapley values, which average a cause's marginal contribution over every ordering and provably still sum to the total.

Construction

We define a counterfactual cost function C(S): the bill if the causes in set S sit at current-quarter levels and the rest stay at last quarter's. The baseline model from R2 evaluates it. With four causes that is sixteen evaluations, memoised.

```
tariff, volume, mix, weather  =  Shapley values over C   (sums to C(all) - C(none))
waste  =  actual_cost - C(all)   (what the model cannot explain)

sum of all five  =  actual_Q3 - actual_Q2   exactly, by construction
```

Additivity is therefore structural. The assertion in our code is a regression guard against floating-point drift and bugs, not a mechanism for forcing the numbers to reconcile.

Where the ranges come from

Annex A requires low and high values. We bootstrap: refit the baseline on 50 resamples, recompute the Shapley split on each, and report the 10th and 90th percentiles. Savings ranges propagate that interval together with our stated assumption ranges for the emission factor and leak persistence. No range is invented.

5. Two guardrails we build first

Nothing is blamed before the causes are removed

Annex A requires tariff, volume, mix and weather effects to be removed before anything is called waste. We enforce this in the type system rather than in a prompt: the leak detector takes the completed bridge as a required argument, so it is impossible to call it earlier. The graph topology is the policy.

Every number is traceable

Each tool result is written to a registry row recording the value, its unit, the tool that produced it and its evidence reference. After the brief is generated we extract every numeral and check it against the registry with a rounding tolerance. Unmatched values trigger one regeneration, then the template fallback. The same registry rows populate evidence_refs in the response and appear in the trace, so one table serves the faithfulness check, the API contract and R12.

6. Data generation, and the circularity risk

We generate our own data to the Annex B schema. That removes the problem of an unverifiable answer key and lets us measure decomposition error exactly, but it introduces a different risk, and we have designed against it deliberately.

> **The risk we are designing against**
>
> If the generator and the analyser are mirror images, a perfect result proves nothing. We therefore generate from explicit physical drivers — furnace from units multiplied by per-product energy intensity, HVAC from cooling degree-days, compressors from shift hours plus a leak beginning on a known date — while the analyser fits its baseline from the CSV files alone and never imports a generator constant. We also give the generator a piecewise temperature response that our linear baseline cannot represent, so a small residual error is expected and honest rather than suppressed.

The generator is seeded and emits a ground_truth.json recording the true split and the location of every injected anomaly. We generate a second holdout set with a different seed and different patterns, and do not run against it until the evaluation pass.

Edge case (Section 4.4)	How we inject it	How we detect it
Sub-meters do not sum to the main meter	Unmetered base load plus per-meter measurement noise	Reconciliation check against a stated tolerance
Missing intervals and meter resets	Drop intervals in blocks; add a cumulative counter rollover	Gap report; reset detected as a negative delta and corrected
Tariff change mid-quarter	Two tariff rows with a boundary inside the period	Price joined per interval, never as a period average

Edge case (Section 4.4)	How we inject it	How we detect it
Mix shift raising energy per unit with no waste	Shift product weights toward a more energy-intense family	Mix term in the bridge absorbs it; waste stays flat
Hot period raising cooling load	Temperature excursion driving cooling degree-days	Weather term in the bridge; HVAC excluded from leak search
Public holiday logged as a normal shift	is_holiday false on a day with near-zero production	Shift calendar versus production cross-check in validation

Each injected case doubles as an evaluation case, and the mix-shift trap is the one we will demo: it is the case where the correct answer blames product mix rather than a machine.

7. Requirements coverage

ID	How we meet it
R1	Loader normalises to the Annex B schema; cleaning resamples to 15 minutes, aligns the four tables on timestamp, repairs gaps, duplicates and meter resets. Validation script runs before any analysis.
R2	Linear baseline predicting energy from volume, product mix and degree-days. R-squared reported in the trace.
R3	Shapley decomposition over the four causes with waste as the residual. Additivity asserted in code.
R4	Tariff joined per 15-minute interval against valid_from and valid_to, so a mid-period price change is handled exactly rather than averaged.
R5	Per sub-meter consumption and share of total, ranked.
R6	Idle load measured as consumption during zero-production intervals, reported by asset with the hours it occurs.
R7	Leak detection from night and weekend baseline persistence, run only after the bridge is complete.
R8	Savings in kWh, money and CO2e, each as a bootstrapped low-high range with the emission factor stated.
R9	Actions ranked by payback period, then by ease.
R10	Brief written by the LLM from the number registry only, with post-generation validation.
R11	Waterfall bill-bridge chart showing the five causes summing to the total.
R12	Every run persists to runs, run_steps and registry_values; /trace/{run_id} returns the Annex C shape.

8. How we will evaluate

Because we generate the data, we know the true split, so decomposition error is measured rather than estimated. Our evaluation set is built from generator ground truth and exceeds the Section 7 minimums: cases where the correct answer blames price or mix rather than machines, numeric checks, out-of-scope questions, what-ifs, and deliberate attempts to make the model invent a figure.

- **Decomposition error** — mean absolute error per cause, in percentage points, against the injected truth.
- **Total reconciliation** — asserted on every run, reported as a pass rate.
- **Waste detection** — injected leak found; no waste claimed on the mix-shift and weather traps.
- **Savings accuracy** — share of cases where the true saving falls inside our reported range.
- **Number faithfulness** — share of briefs passing the registry check unaided, and the count of regenerations and template fallbacks.
- **Latency and cost** — p50 and p95, LLM calls and tokens per request.

Comparing two configurations

For the higher marks offered in Section 7 we will run both Shapley and sequential Laspeyres attribution across 50 generated scenarios and report mean absolute error per cause for each. The choice of method is then justified by numbers rather than by assertion. MOCK_LLM lets us run the whole set repeatedly at no cost.

9. What we have deliberately not built

Not built	Reason
Vector store and retrieval	There is no document corpus in this problem. The brief awards no marks for using one, and it would add a dependency and a failure mode for no capability.
Multi-agent orchestration	Every analytical step is deterministic. A second reasoning agent would introduce non-determinism into a pipeline whose correctness we need to be able to assert.
Machine learning beyond a linear baseline	A gradient-boosted model might fit better, but the baseline has to be explainable in the Q&A and inverted for counterfactuals. We will report the R-squared we achieve and say where it limits us.
Cloud tracing tools	Not permitted; tracing is local, in DuckDB.

10. Questions for our mentor

- Since we generate our own data, will a late data drop still be supplied at 13:45, or should we simulate our own corrected tariff table and sub-meter calibration fix?
- Is an emission factor supplied for CO2e, or do we choose and state our own?
- What reconciliation tolerance is expected between the sub-meter total and the main meter? We plan to state and justify our own.
- For live testing, will judges ingest CSVs in the Annex B schema only, or should the loader tolerate additional or reordered columns? We are building for the latter.

Prepared for the mentor check-in. Architecture, decomposition method and guardrails are settled; implementation is next.
Repository: github.com/silverbolt08/RNG